

MEETHI KAHANI

"Every Cookie Tells A Sweet Story"

SYSTEM ARCHITECTURE GUIDE (AG SPECIFICATION)

Comprehensive Technical Architecture Blueprint: Frontend Modular Engine & Backend Data Systems

Project Name:	Meethi Kahani (D2C Artisanal Cookie E-Commerce)
Lead Tech Architect:	AG (AI Engineering Partner)
Architecture Scope:	Full Stack (React 18 Frontend + Relational Backend DB + Modular CMS + Plugin & Channel Analytics)
Dietary USP & Market:	100% Pure Eggless Artisanal Bakery Mumbai Metropolitan Region (MMR)
Document Status:	Approved AG Architectural Specification v1.1 (Updated)
Primary Color Palette:	Crimson Red (#9A2222), Vanilla Linen (#FAF5EE), Dark Cocoa (#3D2314)
Date of Issuance:	August 2026

ARCHITECTURE PRINCIPLE:

This document provides the definitive architectural masterplan for **Meethi Kahani**. It covers the complete end-to-end blueprint comprising the modular React frontend hierarchy, single master stock database schema, dynamic secret admin/seller portals, comprehensive 12-module analytical suite (including Traffic Acquisition & Plugin Analytics), 1-Day Metro Express logistics model, and security protocols.

1. Executive Summary & System Overview

Meethi Kahani is an ultra-luxury Direct-To-Consumer (D2C) artisanal cookie brand based in Mumbai. The digital platform is engineered to combine authentic heritage recipes with modern digital luxury. It operates on a single-product focus (100% Cookies for Years 1-2) with a strict 100% Pure Eggless guarantee targeting India's massive vegetarian market.

Core System Principles:

- **Strict No-Popups Policy:** Zero intrusive overlays or social proof popups to preserve a serene, high-end luxury shopping experience.
- **Single Master Stock Pool:** All D2C order tracks (Personal, Gifting) draw inventory from a unified central cookie stock table.
- **Secret Portal Architecture:** Admin (`/admin``) and Kitchen Seller (`/seller``) portals are strictly hidden from public navigation and protected by authenticated credentials.
- **Decoupled Analytics Engine:** High-volume sales, traffic channel attribution, and plugin performance analytics are isolated on `/admin/analysis`` to maintain sub-second frontend rendering speed.

2. Frontend Architecture & Modular Component Blueprint

The frontend is built using **React 18 + Vite + JavaScript (ES6+)** and styled with **Pure Vanilla CSS3** (utilizing CSS custom properties for brand design tokens).

7-Module Team Collaboration Map:

To ensure seamless parallel development across a multi-developer team, the application is divided into 7 independent modules inside ``src/modules/``. All modules consume shared contracts from ``StoreContext.jsx`` and ``dataContracts.js``.

Module & Path	Responsible Scope	Key Features & Interfacing
Module 1: HeaderNav <code>`src/modules/HeaderNav/`</code>	Navigation & Global Bar	Centered Logo Header layout (Left: <code>`[Logo]`</code> Drawer, Center: Logo, Right: <code>`[Account & Cart]`</code>). Slide-out navigation drawer.
Module 2: HeroBanner <code>`src/modules/HeroBanner/`</code>	Hero Showcase & Urgency	Dynamic Banner Carousel Slider (Admin-managed) + Freshness Ticker Badge (<code>`[X] Boxes left -- Freshly baked in last 3 hours`</code>).
Module 3: ProductCatalog <code>`src/modules/ProductCatalog/`</code>	Catalog & PDP Details	Category Filter Tabs, Cookie Cards with out-of-stock badges, Dedicated PDP (<code>`/product/:slug`</code>), multi-thumbnail gallery, & 'The Kahani' story modal.
Module 4: CartPayment <code>`src/modules/CartPayment/`</code>	Cart Drawer & Checkout	Slide-Over Cart Drawer, live Cookie Crumb Progress Bar for free delivery, Serviceable Pincode Checker, & Pixel-Perfect Dummy Razorpay Modal.
Module 5: CustomerPortal <code>`src/modules/CustomerPortal/`</code>	Account & Reviews	Sliding Review Carousel (above footer, 1-5 <code>`[Rating]`</code> rating), Customer Account Modal (<code>`[Modal]`</code>), Live Animated Cookie Crumb Order Tracker, & Invoice PDF Download.
Module 6: AdminPortal <code>`src/modules/AdminPortal/`</code>	Super Admin CMS	Secret URL <code>`/admin`</code> , Banner Carousel Manager, Coupons Manager, 2-Level Nested Category & Catalog Manager, Style B GST Invoice Customizer, & Analytics Tab.
Module 7: SellerPortal <code>`src/modules/SellerPortal/`</code>	Bakery Kitchen Operations	Secret URL <code>`/seller`</code> , Live Order Fulfillment Queue (Received -> Baking -> Shipped -> Delivered), Restock (<code>`(+50 Baked)`</code>), & Packaging Stock Tracker.

Multi-Page Routing Hierarchy (SPA):

- ``/`` — Main Storefront (Hero, Offers, Products, Cart Drawer, Footer)

- `/gifting` — Gifting Hub (Build-Your-Own Box, 25% MRP discount, custom wax-sealed card & video QR code)
- `/product/:slug` — Dedicated PDP (Left multi-photo thumbnail gallery, right-side details & Kahani story)
- `/story` — Brand Heritage Narrative Page
- `/admin` — Secret Super Admin Control Center
- `/admin/analysis` — Dedicated Comprehensive Analytics Dashboard (12 Modules)
- `/seller` — Secret Bakery Kitchen Order & Inventory Portal

3. Backend Architecture & Data Models

The backend infrastructure utilizes a highly normalized relational database structure designed for high concurrency, real-time inventory updates, traffic attribution logging, and analytical query execution.

Database Schema Overview (12 Core Relational Tables):

Table Name	Primary / Foreign Keys	Core Purpose & Column Highlights
`Customers`	`id` (PK)	Stores customer profiles, phone, email, saved address JSON, Kahani Coins ('Kisse') balance, and total order count.
`Admin_Sellers`	`id` (PK)	Stores admin & bakery seller login credentials, role (`SUPER_ADMIN`, `SELLER_KITCHEN`), kitchen location, and status.
`Categories`	`id` (PK)	Product categories (`Cookies`, `Fudges`, `Gift Sets`), display order, and active flag.
`Cookie_Stock`	`id` (PK), `category_id` (FK)	Single Master Inventory pool for cookie SKUs. Contains name, slug, price (₹), stock_pieces, photo_urls, and Kahani story text.
`Packaging_Stock`	`id` (PK)	Tracks physical packaging inventory: Box Tins (4, 8, 12 count), Wax Seals, Ribbons, and Thermal Mailers with low-stock alerts.
`Hero_Banners`	`id` (PK)	Stores live homepage carousel banners, headline, subtitle, desktop/mobile image URLs, CTA link, priority sequence, and status.
`Coupons`	`id` (PK)	Admin promo codes, discount_type (`PERCENT`, `FLAT`), discount_val, min_order_val, expiry, and total_usage_count.
`Orders`	`id` (PK), `customer_id` (FK)	Master order ledger. Order type (`PERSONAL`, `GIFTING`), subtotal, discount, final_amount, status, payment_mode, payment_status, shipping address fields, and acquisition traffic source.
`Order_Items`	`id` (PK), `order_id` (FK), `cookie_id` (FK)	Line items per order. Quantity, unit_price, total_price. Auto-decrements `Cookie_Stock` during creation.
`Serviceable_Pincodes`	`id` (PK)	Mumbai MMR 6-digit delivery pincodes, city_name, delivery_tier (`1-Day Express`), and active boolean status.
`Customer_Reviews`	`id` (PK), `customer_id` (FK)	Customer feedback ratings (1-5 ★), review text, admin approval status (`is_approved`), and homepage carousel flag.
`Traffic_Analytics`	`id` (PK), `session_id`	Logs visitor traffic sessions: source (`GOOGLE_ORGANIC`, `GOOGLE_ADS`, `WHATSAPP_CAMPAIGN`, `INSTAGRAM_BIO`, `DIRECT`), UTM params, device, landing_page, and conversion status.

Standardized 6-Field Address Architecture (For Fast SQL Regional Analytics):

To enable 1-click regional grouping and automated shipping courier API payloads, addresses in `Customers` and `Orders` tables are explicitly partitioned into 6 standard fields:

```
`address_line_1` | `address_line_2` | `town` (e.g. Bandra, Vashi, Thane) | `city` (Mumbai) | `pincode` (6-digit) | `state` (Maharashtra)
```

4. Dedicated Comprehensive Analytics Suite (`/admin/analysis`)

To maintain lightning-fast page loads for customers, all heavy data aggregations, SQL window functions, traffic channel attribution, and plugin performance tracking are isolated inside `/admin/analysis`.

Module #	Analytics Module Name	Core Technical & Metrics Functionality
Module #1	Stock Behavior Analytics (TOP PRIORITY)	Single stock depletion velocity Week-to-Week & Month-to-Month baking forecast. Prevents inventory stockouts.
Module #2	Traffic Source & Visitor Attribution (NEW)	Detailed breakdown of unique visitors & sessions by acquisition channel: Google Organic (SEO & JSON-LD recipe rich snippets CTR), Google Ads , WhatsApp Campaigns & Direct Links , Instagram/Meta Bio Links & Ads , & Direct URL traffic. Measures bounce rate & channel conversion rates.
Module #3	Ecosystem Plugins Performance (NEW)	Performance, conversion ROI, & operational health across all installed plugins: WhatsApp Cart Abandonment (carts recovered, revenue saved ■), WhatsApp Business API (delivery rate %, msg open rate), Digital Gift Cards , Kisse Loyalty Coins , Hyper-Local Logistics APIs (dispatch speed, transit mins), & Razorpay/PhonePe Payment Gateway (success rate % per payment mode).
Module #4	Customer Demographics	Mumbai MMR sub-zone sales breakdown (South Mumbai, Western Suburbs, Navi Mumbai, Thane) + Age/Gender demographics.
Module #5	Financial & Revenue	Gross revenue (■), AOV (■), Personal vs Gifting revenue split, and Peak Ordering Hours/Days heatmap.
Module #6	Product Flavor Velocity	Top 5 Best-Selling Cookie SKUs, slow-moving item alerts, and popular Gifting combo selections.
Module #7	Promo & Campaign ROI	Promo code redemption counts & WhatsApp marketing campaign ROI conversion tracking.
Module #8	Customer Retention	Repeat customer rate % and Kahani Coins ('Kisse') balance accumulation vs redemption velocity.
Module #9	Delivery & Logistics	Average delivery duration (hours), on-time delivery rate %, and top order volume pincodes.
Module #10	Cart & Funnel Analytics	Cart abandonment rate % and identification of most abandoned cookie SKUs.
Module #11	Customer Satisfaction	Average store rating (4.9 ■ out of 5) and customer sentiment text keyword analysis.
Module #12	Seller & Kitchen Ops	Cookies baked & restocked per Seller ID, order packing velocity, and batch scrap/waste logs.

Detailed Visitor Channel Attribution Breakdown (Module #2):

- **Google Organic & Rich Snippets:** Tracks sessions originating from Google Search queries. Monitors impression volume and CTR of JSON-LD Schema Recipe cards displaying ■ ratings & pricing directly in Google Search results.
- **WhatsApp Channel Traffic:** Separates visitors arriving from automated WhatsApp order updates, direct customer support links, and broadcast promotional campaigns.
- **Social Media Referrals (Instagram / Meta):** Tracks visitor volume from Instagram bio links, influencer unboxing links, and paid Meta ad campaigns.
- **Direct & Referral Visitors:** Tracks loyal returning customers typing the URL directly (`meethikahani.com`) versus external media coverage.

Plugin Ecosystem Performance Metrics Breakdown (Module #3):

- **WhatsApp Cart Abandonment Plugin:** Abandoned carts detected, 30-min recovery messages dispatched, link click-through %, orders completed, and total recovered revenue (■).
- **Payment Gateway Plugin (Razorpay / PhonePe):** Payment authorization success rate %, failure reasons, UPI QR vs Credit Card vs COD breakdown, modal drop-off rate.
- **Hyper-Local Courier Plugin (Porter / Dunzo):** Rider assignment speed (mins), average delivery duration, delivery failure rate, courier cost per order.
- **Loyalty Program Plugin ('Kisse' Coins):** Active coin balances, redemption frequency, average order value increase for coin redeemers.

5. Operations, Logistics, & Security Architecture

1-Day Metro Express Delivery Logistics Architecture:

- Direct delivery model restricted to Mumbai Metropolitan Region (MMR) to guarantee fresh oven-baked quality.
- Instant pincode validation against `Serviceable\_Pincodes` DB table during cart item addition and checkout.
- Auto-assignment rider integration via hyper-local courier APIs (Porter / Dunzo / Shiprocket).

Packaging Material Inventory Tracking Logic:

The Seller Portal (`/seller`) manages physical packaging assets alongside cookie stock. When a kitchen staff marks an order as 'Packed', the system auto-decrements packaging inventory based on box size:

```
`Box of 4 Tin` (-1) | `Wax Seal Stamp` (-1) | `Branded Satin Ribbon` (-1m) | `Thermal Mailer` (-1)
```

Authentication & Role-Based Access Control (RBAC):

- `SUPER\_ADMIN`: Full access to `/admin` (CMS, Banner Editor, Coupons, Invoice Customizer, & `/admin/analysis`).
- `SELLER\_KITCHEN`: Access restricted to `/seller` (Order fulfillment queue & packaging stock replenishment).
- Secret URLs (`/admin` and `/seller`) are omitted from sitemaps, robots.txt, and public header navigation.

GST Tax Invoice Customizer (Style B Modern Clean Template):

PDF GST Tax Invoices are generated on-the-fly using Style B specification. Downloadable by customers on Checkout Success, Customer Account Order History, and via automated WhatsApp confirmation links.

6. External Integrations & Plugin Ecosystem Architecture

Integration Category	Target Provider / API	Architectural Workflow & Benefit
WhatsApp CRM & Alerts	Interakt / Wati API	Automated WhatsApp order receipts, live delivery tracking links, and targeted CRM blasts.
Cart Drop Recovery	WhatsApp Abandonment Plugin	Triggers automated WhatsApp reminder 30 mins after cart drop-off (recovers 15-25% lost sales).
Payment Gateway	Razorpay / PhonePe SDK	Handles UPI QR code scanning & Card payments with instant webhook status updates to DB.
Hyper-Local Logistics	Porter / Dunzo / Shiprocket	Dispatches riders for 1-Day Metro Express delivery and returns real-time GPS tracking.
Loyalty Program	Kahani Coins ('Kisse') Module	Accrues 1 'Kissa' coin per █10 spent. Auto-applies discount redemptions in Cart Drawer.
SEO & Rich Snippets	Google Recipe & Product Schema	Injects JSON-LD rich snippets to display cookie photos, prices, and █ ratings in Google Search.

7. Risk Mitigation & System Architectural Constraints

The architecture incorporates explicit mitigation protocols for operational and technical edge cases:

Identified Risk Area	Potential Impact	Architectural Mitigation Strategy
----------------------	------------------	-----------------------------------

Gifting Volume Stock Depletion	Surges in B2C Gifting Box purchases draining single cookie stock pool.	System enforces real-time stock availability check in cart drawer and triggers low-stock alerts in Kitchen Seller Portal.
Food Safety & Transit Damage	Strict no-refund food safety policy vs transit damage claims.	Automated 24-hour Damage Replacement Workflow (fresh replacement dispatched upon photo upload, zero cash refunds).
Corporate B2B Complexity Elimination	Multi-address shipping, 30-day PO cycles, and B2B GST tax overhead.	Corporate track removed from Phase 1 scope, simplifying platform strictly to D2C Personal & Gifting purchase flows.

Architectural Approval & Sign-Off:	AG AI Engineering Partner
Document Hash & Verification:	MK-AG-ARCH-V1.1-2026-UPDATED
Confidentiality Notice:	Proprietary System Specification — Meethi Kahani Foods Pvt Ltd